TURKIC INTERNATIONAL MATHEMATICAL OLYMPIAD

TIMO

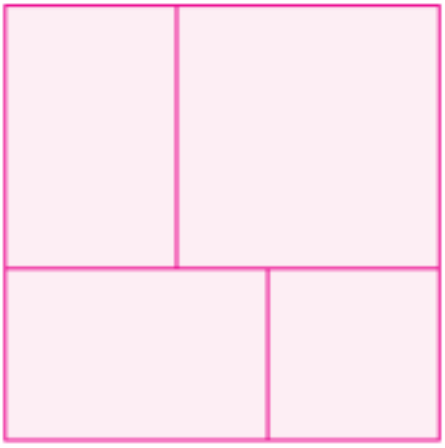
GRADE 7

Mathematical Horizons: Uniting Turkic Minds, Expanding Global Genius.

SCHOLARSHIP FROM TURKISH UNIVERSITIES

MEDIPOL · BAHÇEŞEHİR · İSTANBUL · TOBB · İSTANBUL GALATA UNIVERSİTESİ

1.



In the figure, two different squares and two identical rectangles are drawn inside the large square with area of 225 cm². The long side of one of the rectangles is 3 cm more than the short side.

According to this, what is the difference in the areas of the two squares drawn inside?

A) 54 B) 49 C) 45 D) 42 E) 36

2. What is the result of?

$$(5\cdot 7\cdot 11)\left(\frac{1}{5}-\frac{2}{7}+\frac{3}{11}\right)$$

A) 69 B) 72 C) 75 D) 77 E) 81

3. Erdem, who plays football in an amateur club, controls one of every 4 passes thrown to him and shoots at the goal. One out of every 3 shots finds the goal. On the other hand the goalkeeper can save one out of every two shots.
What is the minimum number of passes that Erdem needs to receive to ensure that he scores at least one goal?

A) 14 B) 16 C) 18 D) 24 E) 28

4. Emre completes a square parkour with a side length of 30 m in 1 minute. Besides , Ersin completes a rectangular parkour with a long side of 100 metres in 3 minutes.
Since the speeds of Emre and Ersin are the same, how many metres is the short side of the rectangular track?

A) 20 B) 30 C) 60 D) 70 E) 80

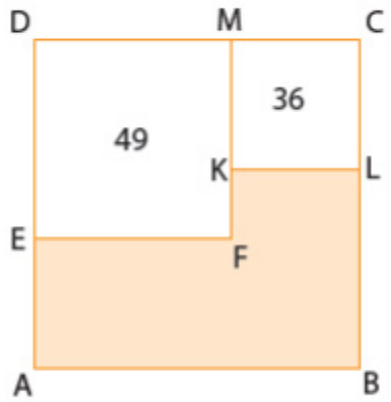
5. Numbers whose straight and reverse readings are the same are called Palindrome Numbers. For example; 131, 41014, etc.

1009735379001 palindrome number is given.

What is the sum of the digits of the number forming the next palindrome number after this number?

A) 46 B) 47 C) 49 D) 51 E) 55

6.



There are 3 squares in the figure above.
Since the area of the smallest square is 36 cm² and the area of the median square is 49 cm², what is the perimeter of the shaded shape?

A) 35 B) 37 C) 40 D) 43 E) 48

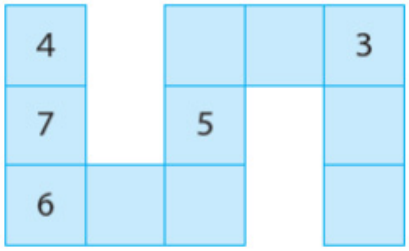
7. The following ratios are given among the numbers a, x, y, and z.

$$\frac{x}{y}=\frac{3}{4} \qquad \frac{x}{z}=\frac{1}{2} \qquad \frac{a}{z}=\frac{5}{6}$$

What is the value of the ratio of $\frac{z^2-y^2}{a\cdot x}$?

A) $\frac{3}{4}$ B) $\frac{2}{5}$ C) 1 D) $\frac{4}{3}$ E) $\frac{5}{2}$

8. In the table, some of whose numbers are deleted, the numbers are written in such a way that the sum of the three numbers written vertically and horizontally is equal to each other.



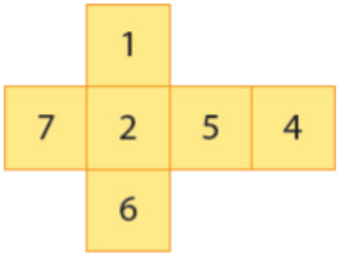
What is the sum of the deletede numbers in the table?

A) 43 B) 39 C) 37 D) 31 E) 28

9. Each of the different numbers obtained by writing each digitr as much as itself will be written side by side and summed as shoen below.
1+22+333+4444+…+999999999
What is the sum of the last two digits of the number obtained by this summation?

A) 12 B) 13 C) 14 D) 15 E) 16

10. The open form of a cube with numbers written on it is given below.When this cube is closed, three faces meet at each corner.



According to this, what is the maximum sum of the numbers written on the three faces that meet at any corner?

A) 14 B) 15 C) 16 D) 17 E) 18

11. The following informations are given for Ayla, Buse, Cem, Deniz and Ece who are sitting around a round table.

- Ece is not sitting next to Deniz
- Buse is sitting in between Ayla and Deniz

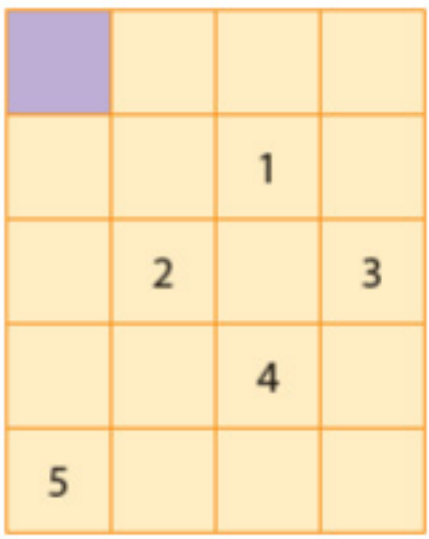
According to this, in which of the following choices are the people to the right and left of Ece given?

A) Ayla-Buse B) Buse-Deniz C) Cem-Deniz D) Ayla-Cem E) Buse-Cem

12. Ahmet and Burak are swimming in a river which is 700 metres long and flows at a constant speed. The swimming speeds of each of Ahmet and Burak are different from each other and the speed of each in the same direction with the current is 2 times their speed against the current. When they start swimming against the current at the same time, Burak finishes 700 metres and when he turns back, he meets Ahmet after swimming 70 metres.
Accordingly, when Burak finishes the race, how many more metres does Ahmet have to swim to finish the race?

A) 350 B) 320 C) 300 D) 280 E) 240

13. A locust that starts jumping from the painted (top left) square in the grid below first jumps 3 squares horizontally or vertically, then jumps two squares perpendicular to this path and makes 1 move.
Without leaving this ground the locust can make as many moves as it wants.



According to this, which of the numbered square is impossible for the locust to reach?

A) 1 B) 2 C) 3 D) 4 E) 5

14. Ahmet wants to make two squares with integer side lengths using a all of the 76 cm long wire. Then he fills the big square with the squares that are identical to the small square he made.
At most how many identical small squares can he fit into the big square?

A) 225 B) 256 C) 289 D) 324 E) 362

15. In a class of 30 students, 3 different books will be distributed to each student. These books belong to maths, physics, chemistry and biology courses. The books of a course are identical in themselves .
Since a total of 25, 24 and 21 maths, physics and chemistry books were distributed respectively, how many students did not receive a biology book?

A) 9 B) 10 C) 11 D) 12 E) 13

16. While Hülya, Hatice and Bilge were studying together, they took a mock test and timed it and stayed in the test for 61, 65 and 53 minutes, respectively.
Since Hatice started the test at the same time as Hülya and finished the test at the same time as Bilge, how many minutes did the three of them take the test at the same time?

A) 41 B) 43 C) 45 D) 47 E) 49

17. What is the maximum product of two prime numbers whose sum is 20?

A) 51 B) 63 C) 75 D) 91 E) 99

18. If the sum A + 19876 is equal to a natural number with different digits, what is the sum of the digits of the smallest positive integer value A can take?

A) 13 B) 15 C) 17 D) 19 E) 21

19. The rule of the number sequence whose first 4 terms are given below is as follows:
From the 5th term onwards, each term is the largest digit different from the previous 4 terms.
9, 7, 6, 2, ...
According to this, what is the 2024th term of this sequence?

A) 8 B) 9 C) 7 D) 6 E) 5

20. Enes can walk from home to the bus stop in 4 minutes. From this bus stop, he can reach the metro station with buses that arrive every 15 minutes for a 20-minute journey and from there he can reach the school with the metro that arrives every 10 minutes for a 30-minute journey.
According to this, which of the following cannot be the time in minutes for Enes to go from home to school?

A) 80 B) 78 C) 64 D) 60 E) 56

21. If a traveller can go from village A to village B on the river bank in 2 hours by motorised boat and from village B to village A in 6 hours by the same boat, how many hours does it take to go from village A to village B on a raft (without motor and oars)?

A) 3 B) 4 C) 5 D) 6 E) 7

22. If Berke buys some erasers at 3 liras each, he has 1 lira left; if he buys some pencils at 7 liras each, he has 2 liras left.
If Berke buys some notebooks at 21 liras each, what is the amount of his remaining money which is less than 21 liras?

A) 16 B) 18 C) 19 D) 17 E) 15

23. Given
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$$A = 10000^{10000} + 1000^{1000} + 100^{100} + 10^{10} + 1$$

A) 1 B) 2 C) 3 D) 4 E) 5

24. How many 7-digit numbers whose sum of the digits is 3 are there?

A) 18 B) 24 C) 28 D) 30 E) 32

25. Given, $A = 98^{100} + 99^{100}$
What is the unit digit of the number A?

A) 2 B) 3 C) 5 D) 7 E) 8